

Geospatial Python Mastery

A professional beginner-to-hero course outline focused on Python, spatial data, PostgreSQL/PostGIS, and CityJSON workflows

Course Level	Beginner to Advanced Practitioner
Duration	10 weeks / 40 contact hours / 60+ hours guided practice
Format	Instructor-led lectures, labs, weekly assignments, and a capstone project
Primary Stack	Python, GeoPandas, Shapely, PyProj, Rasterio, PostgreSQL, PostGIS, CityJSON

Program purpose

This program takes a student from core Python literacy to production-ready geospatial engineering. It is structured to build strong data-handling fundamentals first, then move into spatial computation, database-backed workflows, and city-scale 3D data parsing and validation.

Graduate profile

By the end of the course, a learner can ingest heterogeneous geospatial data, transform coordinates, run spatial analyses in Python, connect securely to PostgreSQL/PostGIS, author efficient spatial queries, and parse CityJSON datasets for practical urban analytics.

1. Learning outcomes

- Write clean Python code for geospatial data workflows, including functions, modules, classes, and virtual environments.
- Manipulate JSON and file-based datasets with reliable validation, logging, and error handling.
- Use GeoPandas, Shapely, and PyProj for vector operations, coordinate reference systems, and geometry transformations.
- Perform spatial database operations in PostgreSQL/PostGIS, including indexing, joins, buffering, intersection, and aggregation.
- Parse, inspect, and normalize CityJSON structures for 3D city-model use cases.
- Deliver a portfolio-grade capstone that combines file processing, spatial analytics, and database integration.

2. Audience and prerequisites

Target audience: analysts, GIS learners, software developers, data engineers, and technical staff entering geospatial work.

Prerequisites: basic computer literacy. No prior GIS background is required, but comfort with algebra and structured thinking is helpful.

3. Tools and environment

- Python 3.11+ with pip and virtual environments
- JupyterLab or VS Code
- GeoPandas, Shapely, PyProj, Fiona, Rasterio, Pydantic, Requests, Psycopg / SQLAlchemy
- PostgreSQL 15+ with PostGIS extension
- QGIS for visual verification and QA
- CityJSON sample datasets and validation utilities

4. 10-week training timeline

Week	Theme	Core topics	Practice outcome
Week 1	Orientation and Python fundamentals	Set up the environment, learn syntax, data types, control flow, functions, and debugging.	Mini-lab: build a reusable data-cleaning script.
Week 2	Python for data work	Lists, dictionaries, comprehensions, modules, packages, exceptions, logging, and testing basics.	Assignment: create a robust JSON parser with validation.
Week 3	JSON and file processing	Read/write JSON, CSV, TXT, GeoJSON, newline-delimited data, path handling, encoding, and batch processing.	Lab: normalize a folder of mixed geospatial files.
Week 4	Geospatial foundations	Coordinates, projections, CRS theory, geometry types, topology, spatial reference management, and file formats.	Exercise: compare WGS84 and projected workflows.

Week	Theme	Core topics	Practice outcome
Week 5	Spatial libraries in Python	GeoPandas, Shapely, PyProj, Fiona, and Rasterio; vector operations, joins, overlays, measurements, and raster intro.	Lab: analyze points, lines, and polygons for a local study area.
Week 6	PostgreSQL connection and data loading	Connect Python to PostgreSQL, use SQLAlchemy/PyScopg, create schemas, load tables, and manage transactions.	Assignment: build an ETL loader for spatial records.
Week 7	PostGIS core queries	Spatial types, spatial indexes, ST_Intersects, ST_Within, ST_Buffer, ST_Area, ST_DWithin, ST_Transform, and query planning.	Lab: answer urban planning questions using PostGIS.
Week 8	Advanced spatial analytics	Spatial joins at scale, performance tuning, aggregation, nearest-neighbor logic, validity checks, and geometry repair.	Exercise: optimize slow spatial queries and indexes.
Week 9	CityJSON parsing and 3D geospatial data	CityJSON schema, object hierarchies, semantics, geometry arrays, transformations, and parsing workflows in Python.	Lab: extract building footprints and metadata from CityJSON.
Week 10	Capstone project and presentation	Design, implement, test, and present a complete geospatial solution that combines files, Python, and PostGIS.	Final deliverable: notebook, script package, SQL, and short presentation.

5. Module-by-module syllabus

Module A - Python foundations

- Syntax, indentation, variables, types, control flow, functions, and script structure
- Modules, packages, virtual environments, and dependency management
- Error handling, assertions, logging, and debugging discipline
- Code style, documentation, and test-first habits

Module B - JSON and file processing

- JSON serialization and deserialization
- Schema awareness, data validation, and robust parsing
- Working with CSV, TSV, TXT, NDJSON, GeoJSON, and file trees
- Batch processing, path management, and safe file I/O

Module C - Spatial Python stack

- GeoPandas data frames and spatial joins
- Shapely geometry operations and topology
- PyProj coordinate transformations and CRS management
- Raster basics with Rasterio where raster-vector integration is required

Module D - PostgreSQL and PostGIS

- Database setup, schema design, and secure connectivity from Python
- Loading spatial data and managing transactions
- Spatial SQL, indexing, and query optimization
- Production patterns for analytical workflows and repeatable jobs

Module E - CityJSON and 3D city models

- CityJSON structure, semantics, and geometry representations
- Parsing, transformation, and feature extraction in Python
- Quality checks, conversion thinking, and downstream analytics use cases
- Bridging 3D city objects to 2D GIS and database workflows

6. Assessments and grading model

- Weekly lab exercises - 30%
- Two short quizzes - 10%
- Mid-course practical - 20%
- Capstone project - 35%
- Professionalism, code quality, and participation - 5%

7. Capstone project options

- Urban accessibility analyzer: compute service areas, nearest facilities, and underserved neighborhoods.
- Spatial ETL pipeline: ingest file-based geospatial data into PostGIS with validation and logging.
- CityJSON insight tool: parse 3D city objects and produce summary metrics and derived GIS layers.
- Land-use intelligence dashboard: combine spatial queries, exports, and map-ready outputs.

8. Recommended learner deliverables

- Clean Git repository with source code and instructions
- Annotated notebooks or scripts for each major module
- SQL scripts for PostGIS setup and analysis
- Final report describing methodology, results, and limitations
- Presentation slides or a recorded demo for stakeholder review

9. Teaching philosophy

The course is designed around progressive mastery: each week reinforces the previous week while adding one new layer of geospatial complexity. Students move from code literacy to spatial reasoning, then to production-grade database and 3D data workflows. Every topic should be grounded in a practical dataset and a real business or civic use case.